ICT Class 8 - Formative Assessment 1

Formative Assessment (FA-1)

- Class 8

Class: Class 8

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Functions

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output of the following Python code?

```
def greet(name):
    return "Hello, " + name

print(greet("Alice"))
```

a. “Hello, Alice”
b. “Hello, “
c. “Alice”
d. Error

b. Which keyword is used to define a function in Python?
c. `func`
d. `define`
e. `def`
f. `function`

c. A function that returns a value is called a:
d. Void function
e. Pure function
f. Generator function
g. Lambda function

d. What is the default return value of a Python function that does not have a `return` statement?
e. 0
f. False
g. `None`
h. Empty string

e. In Python, which of the following is used to call a function?
f. `()`
g. `[]`
h. `{}`

Section II: Short Answer (5 x 2 = 10 marks)

a. Define a function. Give its syntax in Python.

b. Differentiate between **arguments** and **parameters** with one example each.

c. Write the output of:

```
def factorial(n):
    if n == 0:
        return 1
    return n * factorial(n - 1)

print(factorial(5))
```

d. Explain the difference between a **local variable** and a **global variable** inside a function.

e. What is the purpose of the `return` statement in a function? Give an example.

Section III: Long Answer (5 x 1 = 5 marks)

a. Write a Python function `is_prime(n)` that checks whether a given number `n` is a prime number or not. Include docstring, proper indentation, and handle edge cases. Call the function for 5 test values and show the output.

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Defines a function `calculate_bmi(weight_kg, height_m)` that computes Body Mass Index
- Includes input validation (positive numbers only)
- Categorises the BMI: Underweight (< 18.5), Normal (18.5-24.9), Overweight (25-29.9), Obese (>= 30)
- Calls the function for at least 4 different persons and displays a formatted table of results

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)

b. Completion of all assigned classwork and homework (2 marks)

c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks

Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Formative Assessment 2

Formative Assessment (FA-2)

- Class 8

Class: Class 8

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. Which of the following creates an empty list in Python?
b. `my_list = ()`
c. `my_list = {}`
d. `my_list = []`
e. `my_list = dict()`
b. What is the time complexity of accessing an element in a Python list by index?
c. O(n)
d. O(1)
e. O(log n)
f. $O(n^2)$
c. Which of the following is **immutable** in Python?
d. List
e. Dictionary
f. Tuple
g. Set
d. What will be the output of:

```
d = {"a": 1, "b": 2}
d["c"] = 3
print(len(d))
```

a. 2
b. 3
c. Error
d. None
e. Which method adds an element to the end of a list?
f. `add()`
g. `insert()`
h. `append()`
i. `extend()`

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between a **list** and a **tuple**. Give two use cases of each.

b. Write the output of:

```
my_list = [10, 20, 30, 40, 50]
print(my_list[1:4])
print(my_list[::-1])
```

c. Explain the difference between `append()` and `extend()` with examples.

d. Write a Python code snippet to iterate over all key-value pairs in a dictionary using the `items()` method.

e. What is a **nested dictionary**? Give one real-life example where it is useful.

Section III: Long Answer (5 x 1 = 5 marks)

a. Write a Python program that manages a student marks database using a dictionary of lists. The program should:

- Store marks for at least 5 subjects for 3 students
- Calculate the average mark per student
- Find the highest and lowest average
- Display the results in a formatted table

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Creates a list of 10 integers entered by the user
- Uses list methods to: sort, reverse, find the maximum, find the minimum
- Computes the sum and average without using built-in `sum()` or `len()`
- Creates a tuple from the sorted list and displays both

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)

b. Completion of all assigned classwork and homework (2 marks)

c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
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Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Formative Assessment 3

Formative Assessment (FA-3)

- Class 8

Class: Class 8

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL - Database Fundamentals

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. Which SQL command is used to retrieve data from a database?
b. `INSERT`
c. `SELECT`
d. `UPDATE`
e. `DELETE`
b. Which clause is used to filter rows in a SQL query?
c. `GROUP BY`
d. `ORDER BY`
e. `WHERE`
f. `HAVING`
c. The SQL command `DELETE FROM students WHERE roll_no = 5;` will:
d. Delete the entire `students` table
e. Delete all rows from `students`
f. Delete only the row where `roll_no = 5`
g. Delete only the `roll_no` column
d. Which data type is most suitable for storing a student's name in SQL?
e. `INT`
f. `DATE`
g. `VARCHAR`
h. `BOOLEAN`
e. What does the `DROP TABLE` command do?
f. Removes all data but keeps the table structure
g. Removes the table structure and all data
h. Removes only duplicate rows
i. Removes all indexes

Section II: Short Answer (5 x 2 = 10 marks)

a. Write the SQL command to create a table `Books` with columns: `book_id` (INT, Primary Key), `title` (VARCHAR 100), `author` (VARCHAR 50), `price` (DECIMAL), `stock` (INT).
b. Differentiate between `DELETE`, `TRUNCATE`, and `DROP` commands.
c. Write SQL queries to:
d. Insert 3 records into the `Books` table
e. Update the price of a book where `book_id = 2`
f. Delete a book where `stock = 0`
d. What is a **Primary Key**? What are its constraints?
e. Write a SQL query to display all books from the `Books` table sorted by price in descending order, and only show books priced above ₹500.

Section III: Long Answer (5 x 1 = 5 marks)

a. You are given a table `Students` with columns: `roll_no`, `name`, `class`, `section`, `marks`, `grade`. Write SQL queries for:
b. Display students who scored more than 80 marks
c. Count the number of students in each section
d. Find the maximum, minimum, and average marks
e. Display students sorted by marks in descending order
f. Show only the top 5 students

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Connects to an SQLite database (in-memory is acceptable)
- Creates a table `Employees` with columns: `emp_id`, `name`, `department`, `salary`, `joining_date`
- Inserts at least 8 records using user input or predefined data
- Performs 5 different SQL operations: SELECT with WHERE, ORDER BY, GROUP BY, aggregate functions, and UPDATE
- Displays results of each query in a formatted manner

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)
b. Completion of all assigned classwork and homework (2 marks)
c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Formative Assessment 4

Formative Assessment (FA-4)

- Class 8

Class: Class 8

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: HTML and CSS

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. Which HTML tag is used to create a hyperlink?
b. `<link>`
c. `<a>`
d. `<href>`
e. `<url>`
b. Which CSS property is used to change the text colour?
c. `text-color`
d. `font-color`
e. `color`
f. `foreground`
c. Which HTML tag is used to display an image?
d. `<image>`
e. `<pic>`
f. `<img>`
g. `<src>`
d. Which CSS selector targets an element with the class name “header”?
e. `#header`
f. `.header`
g. `header`
h. `*header`
e. Which of the following is a **block-level** element in HTML?
f. `<span>`
g. `<a>`
h. `<div>`
i. `<img>`

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between HTML and CSS. Explain their roles in web development.

b. Write the HTML code to create a table with 3 columns (Name, Age, Grade) and 4 rows of student data.
c. What is the difference between **inline CSS**, **internal CSS**, and **external CSS**? Give one example of each.
d. Explain the CSS box model with a labelled diagram.
e. Write the CSS to:
f. Set the background colour of the body to `#f0f0f0`
g. Add a 2px solid blue border to all `<h2>` elements
h. Set the font size of paragraphs to 16px with line height 1.5

Section III: Long Answer (5 x 1 = 5 marks)

a. Create a complete HTML page for a **School Library Portal** that includes:
b. A navigation bar with 4 links (Home, Catalogue, About, Contact)
c. A header with school name and logo placeholder
d. A table listing 6 books with columns: Title, Author, Genre, Available
e. A form with input fields for student name and book search
f. CSS styling including: colour scheme, hover effects on links, styled table with alternating row colours, responsive form layout

Part B: Hands-on Practical Lab Task (5 marks)

Create an HTML page for a **Personal Portfolio** that:

- Has a header with your name and a tagline
- Includes an “About Me” section with a paragraph and an image placeholder
- Has a skills section displaying at least 5 skills in a grid layout using CSS
- Contains a contact form with fields: Name, Email, Message, and a Submit button
- Uses external CSS with proper selectors and responsive design principles

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)
b. Completion of all assigned classwork and homework (2 marks)
c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)

Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Practical Lab Exam 1

Practical Lab Exam 1

- Class 8

Class: Class 8

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Python Functions and Data Structures

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
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Lab Tasks (15 marks)

Task 1: Function-Based Calculator (5 marks)

Write a Python program that:

- Defines functions `add(a, b)`, `subtract(a, b)`, `multiply(a, b)`, `divide(a, b)`
- Handles division by zero with proper error messaging
- Uses a `menu-driven` approach where the user selects an operation
- Calls the appropriate function based on user input
- Repeats until the user chooses to exit
- Displays results with proper formatting

Task 2: List Operations Manager (5 marks)

Write a Python program that:

- Creates a list of at least 15 student names (user input or predefined)
- Implements a menu with options: Add, Remove, Search, Sort, Reverse, Display All
- Uses appropriate list methods for each operation
- Includes input validation (e.g., cannot remove a non-existent name)
- Displays the list after each operation

Task 3: Dictionary-Based Marks Manager (5 marks)

Write a Python program that:

- Creates a dictionary to store marks for 8 students in subjects: Math, Science, English, SST
- Allows the user to:
  - Add a new student with marks
  - Update marks for an existing student
  - Display all students and their marks
  - Calculate and display the average per student
  - Find the topper (highest average)
- Uses nested dictionaries and proper formatting for output

Viva Voce (5 marks)

1. What is the difference between a function definition and a function call? (1 mark)

2. Can a function have both default parameters and variable-length arguments? Give an example. (1 mark)
3. Why are dictionaries preferred over lists for storing key-value pairs? (1 mark)
4. What is the difference between `remove()` and `pop()` for lists? When would you use each? (1 mark)
5. How does Python handle memory when you create a deep copy vs a shallow copy of a nested list? (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Practical Lab Exam 2

Practical Lab Exam 2

- Class 8

Class: Class 8

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: SQL with Python and Web Development

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Task 1: SQL Database Operations with Python (5 marks)

Write a Python program using SQLite that:

- Creates a database `school.db` with a table `Students` containing:
  - `roll_no` (INTEGER PRIMARY KEY)
  - `name` (TEXT NOT NULL)
  - `class` (TEXT)
  - `section` (TEXT)
  - `marks` (INTEGER)
- Inserts at least 8 records
- Performs and displays results of:
  - SELECT with WHERE clause
  - SELECT with ORDER BY
  - UPDATE a student's marks
  - DELETE a student record
  - COUNT and AVG aggregate queries

Task 2: HTML and CSS Web Page (5 marks)

Create a complete HTML page `student_portal.html` with an external CSS file `styles.css` that includes:

- A styled header with school name and navigation links
- A table displaying 6 students with columns: Name, Class, Section, Marks, Grade
- The table should have:
  - Alternating row colours
  - Hover effect on rows
  - Proper borders and spacing
- A styled footer with copyright information
- CSS should use flexbox for layout and responsive design

Task 3: SQL Query Challenge (5 marks)

Given the following database schema:

```
Employees(emp_id, name, department, salary, joining_date)
Projects(project_id, project_name, emp_id, start_date, end_date)
```

Write Python code that:

- Creates both tables with proper relationships (Foreign Key)
- Inserts at least 6 employees and 10 projects
- Executes and displays the results of:
  - Find all employees who earn more than the average salary
  - List departments with more than 2 employees
  - Find employees working on the most projects
  - Calculate total salary expenditure per department
  - Find projects that are still ongoing (no end date or end date > today)

Viva Voce (5 marks)

1. What is the difference between `INNER JOIN` and `LEFT JOIN`? Give one example of each. (1 mark)
2. How does CSS specificity work? Which rule takes priority: inline, internal, or external CSS? (1 mark)
3. What is the purpose of a Foreign Key in SQL? How does it maintain data integrity? (1 mark)
4. What is the difference between `WHERE` and `HAVING` in SQL? When would you use each? (1 mark)
5. Explain the difference between responsive design and adaptive design in web development. (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Summative Assessment 1

Summative Assessment (SA-1)

- Class 8

Class: Class 8

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Functions, Data Structures, SQL, HTML/CSS

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. Which keyword defines a function in Python?

b. func

c. def

d. function

e. define

b. What is the output of len([1, [2, 3], 4])?

c. 4

d. 3

e. Error

f. 5

c. Which SQL command deletes all rows but keeps the table structure?

d. DELETE

e. DROP

f. TRUNCATE

g. REMOVE

d. In HTML, which tag creates a table row?

e. <tr>

f. <td>

g. <th>

h. <row>

e. Which CSS selector has the highest specificity?

f. Class selector .box

g. Element selector p

h. ID selector #main

i. Universal selector *

f. What will be the output?

```
def f(x, y=[]):
    y.append(x)
    return y
```

```
print(f(1))
print(f(2))
```

a. `[1]` then `[2]`
b. `[1]` then `[1, 2]`
c. `[1]` then `[1]`
d. Error
g. Which method returns a new sorted list without modifying the original?
h. `sort()`
i. `sorted()`
j. `reverse()`
k. `reversed()`
h. Which SQL query displays only unique records?
i. `SELECT * FROM table;`
j. `SELECT UNIQUE * FROM table;`
k. `SELECT DISTINCT * FROM table;`
l. `SELECT DIFFERENT * FROM table;`
i. Which CSS property adds space between the element border and its content?
j. `margin`
k. `padding`
l. `border-spacing`
m. `gap`
j. Which of the following is a mutable data structure in Python?
k. Tuple
l. String
m. List
n. Integer

Part B: Short Answer (8 x 2 = 16 marks)

a. Write a Python function `reverse_string(s)` that returns the reversed version of string `s`. Include proper handling for empty strings.
b. Write the output of the following and explain line by line:

```
def factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)

print(factorial(0))
print(factorial(5))
print(factorial(3))
```

c. Differentiate between `list.append()` and `list.extend()` with examples. What happens when you append a list to another list?
d. Write a Python program to count the frequency of each word in the sentence `"the cat sat on the mat the cat"` using a dictionary. Display the result.
e. Write SQL queries for a table `Employees(emp_id, name, department, salary, joining_date)`:
f. Find employees with salary between 30000 and 50000
g. Count employees per department
h. Find the employee with the maximum salary
i. Delete employees who joined before 2020
f. Differentiate between `DELETE` and `TRUNCATE` with three points each.
g. Write the HTML code for a form with: name, email, date of birth (date input), gender (radio buttons), and a submit button. Include proper labels.
h. Write CSS rules to:
i. Create a responsive navigation bar with flexbox
j. Style links with hover underline effect
k. Set a media query for screens below 768px width

Part C: Long Answer (4 x 3 = 12 marks)

a. Write a complete Python program that uses functions and lists to manage a shopping cart. The program should:
- Allow adding items with name, price, and quantity
- Remove items by name
- Calculate the total bill with 18% GST
- Display a formatted receipt with itemised details

b. Write SQL queries for a `School` database with tables: `Students(roll, name, class, section)`, `Subjects(sub_id, sub_name, max_marks)`, `Marks(roll, sub_id, marks_obtained)`:
c. List all students who scored above 80 in any subject
d. Find the percentage of each student (total marks obtained / total max marks)
e. Find the subject with the highest average score
f. List students ranked by total marks descending
c. Create a complete HTML page for a **Student Report Card** that:
- Has a header with school name and logo placeholder
- Includes a styled table with columns: Subject, Marks Obtained, Grade
- Shows summary statistics (Total, Percentage, Class Rank)
- Uses CSS for professional formatting with borders, colours, and spacing
- Has a print-friendly layout

d. Write a Python program using dictionaries and functions to create a simple **Quiz Application** that:
- Stores at least 5 questions with 4 options and correct answers

- Presents questions to the user one by one
- Accepts user input and validates it
- Keeps track of score and displays final results with percentage and grade

Part D: Application Based (2 x 1 = 2 marks)

a. A student writes the following code and gets unexpected output:

```
def add_item(item, cart=[]):
    cart.append(item)
    return cart

cart1 = add_item("apple")
cart2 = add_item("banana")
print(cart1)
print(cart2)
```

Identify the bug, explain why it occurs, and write the corrected version.

b. Read the following SQL query and answer:

```
SELECT department, COUNT(*) as emp_count, AVG(salary) as avg_salary
FROM Employees
WHERE joining_date > '2020-01-01'
GROUP BY department
HAVING AVG(salary) > 40000
ORDER BY avg_salary DESC;
```

Explain what this query does step by step and what each clause's role is.

Part E: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently

Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part F: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Summative Assessment 2

Summative Assessment (SA-2)

- Class 8

Class: Class 8

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Algorithms, Emerging Technologies, Cyber Safety, Computational Thinking

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. Which sorting algorithm has the best average-case time complexity?
b. Bubble Sort
c. Selection Sort
d. Merge Sort
e. Insertion Sort
b. What is the time complexity of binary search on a sorted array of `n` elements?
c. O(n)
d. O(n log n)
e. O(log n)
f. O(1)
c. Which of the following is NOT a characteristics of a good algorithm?
d. Finiteness
e. Ambiguity
f. Definiteness
g. Effectiveness
d. What does IoT stand for?
e. Internet of Technologies
f. Internet of Things
g. Integration of Things
h. Interface of Technology
e. Which technology is the foundation of cryptocurrency?
f. Cloud Computing
g. Artificial Intelligence
h. Blockchain
i. Big Data
f. What is the primary purpose of a flowchart?
g. To write code
h. To represent an algorithm visually

i. To compile programs
j. To store data
g. Which of these is a characteristic of cloud computing?
h. Local storage only
i. On-demand self-service
j. Single-user access
k. Fixed capacity
h. In pseudocode, which symbol represents a decision/condition?
i. Rectangle
j. Diamond
k. Oval
l. Parallelogram
i. What is machine learning?
j. A type of hardware
k. A subset of AI that enables systems to learn from data
l. A programming language
m. A database management system
j. Which cyber safety practice is most important?
k. Sharing passwords with friends
l. Using public Wi-Fi for banking
m. Enabling two-factor authentication
n. Clicking on unknown email links

Part B: Short Answer (8 x 2 = 16 marks)

a. Write an algorithm (in pseudocode) to find the largest element in an unsorted list of `n` numbers. Analyse its time complexity.
b. Differentiate between **artificial intelligence**, **machine learning**, and **deep learning** with one example of each.
c. Explain the concept of **divide and conquer** with an example. How does Merge Sort use this strategy?
d. What is **Big Data**? Explain the 5 V's of Big Data with one sentence each.
e. Write a Python program that implements **linear search** and **binary search** on a list. Compare their performance.
f. Explain three cyber safety measures that every student should follow while using the internet. Why is each important?
g. What is **cloud computing**? Differentiate between IaaS, PaaS, and SaaS with one real-life example each.
h. Write the pseudocode for a **bubble sort** algorithm. Trace it on the list `[5, 3, 8, 1, 2]` showing each pass.

Part C: Long Answer (4 x 3 = 12 marks)

a. Write a complete Python program that implements **Insertion Sort**. Trace the algorithm on the list `[12, 11, 13, 5, 6]` showing each step. Analyse the time complexity for best, average, and worst cases.

b. Explain the following emerging technologies with detailed examples:

- **Blockchain**: Structure, how it works, and one application beyond cryptocurrency
- **Internet of Things (IoT)**: Components, working, and one smart city application
- **Artificial Intelligence**: Two real-world applications in education and healthcare

c. Design a **traffic management system** using algorithms. Write:

- The problem statement
- An algorithm (in pseudocode) that assigns green signal duration based on vehicle density
- A flowchart description for the decision-making process
- How IoT sensors could enhance this system

d. A school wants to digitise its library system. Write:

- A detailed algorithm to search for a book by title, author, or ISBN
- The database design (tables with columns) using SQL
- Python pseudocode for the search function with error handling
- How cloud computing could be used to make the system accessible from anywhere

Part D: Application Based (2 x 1 = 2 marks)

a. A social media company wants to analyse user behaviour. Explain how:

- Big Data analytics can help understand user preferences
- Machine Learning can improve content recommendations
- AI can detect and remove harmful content automatically

b. A farmer wants to use technology to improve crop yield. Explain how:

- IoT sensors can monitor soil moisture and temperature
- Data analytics can predict optimal planting times
- Cloud computing can help access weather data and market prices remotely

Part E: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks

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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part F: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Unit Test 1

Unit Test 1

- Class 8

Class: Class 8

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Functions

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. Which of the following is the correct way to define a function in Python?

b. `function myFunc():`

c. `def myFunc():`

d. `define myFunc():`

e. `func myFunc():`

b. What will be the output?

```
def add(a, b):
    return a + b

print(add(3))
```

a. 3

b. `None`

c. TypeError

d. 0

c. Which of the following creates a **lambda** function?

d. `lambda x: x * 2`

e. `def lambda(x): x * 2`

f. `lambda = function(x): x * 2`

g. `x -> x * 2`

d. What is the output?

```
def greet(name="World"):
    return "Hello, " + name

print(greet())
```

a. “Hello, “

b. “Hello, World”

c. Error

d. `None`

e. Which keyword is used to return a value from a function?

f. `exit`

g. `end`
h. `return`
i. `yield`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write a Python function `power(base, exp)` that returns `base` raised to the power `exp` without using the `**` operator. Call it with `power(2, 5)` and show the output.
b. What is the output of the following? Explain why.

```
x = 10
def modify():
    global x
    x = x + 5
    return x

print(modify())
print(x)
```

c. Differentiate between **recursion** and **iteration**. Give one example of each.
d. Write the output of:

```
def mystery(s):
    return s[::-1]

print(mystery("HELLO"))
print(mystery("PYTHON"))
```

e. Explain what happens when a function does not have a `return` statement. Give a small example.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Unit Test 2

Unit Test 2

- Class 8

Class: Class 8

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. Which method removes and returns the last element of a list?
b. `pop()`
c. `remove()`
d. `del`
e. `discard()`

b. What will be the output?

```
t = (1, 2, 3)
t[0] = 5
```

a. `(5, 2, 3)`
b. `(1, 2, 3)`
c. `TypeError`
d. `ValueError`

c. Which of the following is valid for a dictionary?
d. `d = {[1]: "a", [2]: "b"}`
e. `d = {(1): "a", (2): "b"}`
f. `d = {1: "a", 2: "b"}`
g. `d = {"1", "a", "2", "b"}`

d. What does `dict.keys()` return?
e. A list of values
f. A list of key-value pairs
g. A view object of all keys
h. A tuple of keys

e. List comprehension to create a list of squares of numbers from 1 to 5:
f. `[x**2 for x in range(1,6)]`
g. `[x^2 for x in range(1,5)]`
h. `[square(x) for x in range(1,6)]`
i. `[x*2 for x in range(1,6)]`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write the output of:

```
nums = [10, 20, 30, 40, 50]
del nums[1:3]
print(nums)
```

Explain what happened.

b. Write a Python program to merge two dictionaries `d1 = {"a": 1, "b": 2}` and `d2 = {"c": 3, "d": 4}` and print the result.

c. What will be the output and why?

```
lst = [[1,2],[3,4],[5,6]]
for row in lst:
    print(row[1], end=" ")
```

d. Differentiate between `list.remove(x)` and `list.pop(index)`.

e. Write a program that takes a string and counts the frequency of each character using a dictionary. Display the result sorted by frequency.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 8 - Unit Test 3

Unit Test 3

- Class 8

Class: Class 8

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: SQL - Database Fundamentals

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. Which SQL aggregate function returns the total number of rows?
b. `SUM()`
c. `COUNT()`
d. `TOTAL()`
e. `ROWS()`
b. Which clause is used with aggregate functions to filter groups?
c. `WHERE`
d. `GROUP BY`
e. `HAVING`
f. `ORDER BY`
c. What is the result of `SELECT DISTINCT department FROM Employees;`?
d. Returns all rows including duplicates
e. Returns unique department values only
f. Returns the first row only
g. Error
d. Which command is used to modify an existing table structure?
e. `ALTER TABLE`
f. `UPDATE TABLE`
g. `MODIFY TABLE`
h. `CHANGE TABLE`
e. Which of these is a valid SQL query?
f. `SELECT * FROM Students WHERE marks > 80 AND grade = 'A';`
g. `SELECT * Students WHERE marks > 80 AND grade = 'A';`
h. `GET * FROM Students WHERE marks > 80 AND grade = 'A';`
i. `SELECT * FROM Students IF marks > 80 AND grade = 'A';`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write a SQL query to create a table `Products` with columns: `id` (INT PRIMARY KEY), `name` (VARCHAR 50), `price` (DECIMAL), `quantity` (INT).
b. Write SQL queries to insert 3 records into the `Products` table and update the price where `id = 2`.
c. What is the difference between `WHERE` and `HAVING`? Give one example of each.
d. Write a query to find the average price of all products grouped by `quantity`.
e. Write a query to display products sorted by price in descending order, limiting results to the top 3.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

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Student Name: ***Date:***
Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 8 - Unit Test 4

Unit Test 4

- Class 8

Class: Class 8

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: HTML and CSS

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. Which tag is used to define the largest heading in HTML?
b. `<heading>`
c. `<h6>`
d. `<h1>`
e. `<head>`
b. Which CSS property controls the size of text?
c. `text-style`
d. `font-size`
e. `text-size`
f. `font-style`
c. What does CSS stand for?
d. Creative Style Sheets
e. Cascading Style Sheets
f. Computer Style Sheets
g. Colorful Style Sheets
d. Which HTML attribute is used to specify inline styles?
e. `class`
f. `styles`
g. `style`
h. `font`
e. Which property is used to change the background colour?
f. `bgcolor`
g. `background-color`
h. `color`
i. `bg-color`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write the HTML code to create an unordered list with 4 items and apply a class `menu-list` to it.
b. Explain the difference between `id` and `class` selectors in CSS with examples.
c. Write CSS to style a button with: padding 10px 20px, background blue, white text, rounded corners (5px), and a hover effect that changes background to dark blue.
d. What is the purpose of the `<!DOCTYPE html>` declaration?
e. Write HTML and CSS code to create a card layout with a title, image placeholder, description text, and a “Read More” link.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

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- Real-life connections: “Where I see this outside school” (1 mark)